2.08 Sequences and Series: Geometric Sequences

Worked Solutions 1–72

1 a)

$$a = 3, \quad r = \frac{12}{3} = 4$$
$$u_n = 3 \times (4)^{n-1}.$$

1 b)

$$a = 80, \quad r = \frac{120}{80} = \frac{3}{2}$$
$$u_n = 80 \times (\frac{3}{2})^{n-1}.$$

1 c)

$$a = 240, \quad r = \frac{120}{240} = \frac{1}{2}$$
$$u_n = 240 \times (\frac{1}{2})^{n-1}.$$

1 d)

$$a = 10, \quad r = \frac{9}{10}$$
$$u_n = 10 \times (\frac{9}{10})^{n-1}.$$

1 e)

$$a = 22, \quad r = -\frac{44}{22} = -2$$
$$u_n = 22 \times (-2)^{n-1}.$$

1 f)

$$a = \frac{1}{3}, \quad r = \frac{-4/3}{1/3} = -4$$
$$u_n = \frac{1}{3} \times (-4)^{n-1}.$$

1 g)

$$a = \frac{9}{10}, \quad r = \frac{-1}{9/10} = -\frac{10}{9}$$
$$u_n = \frac{9}{10} \times (-\frac{10}{9})^{n-1}.$$

1 h)

$$a = 88, \quad r = -\frac{66}{88} = -\frac{3}{4}$$
$$u_n = 88 \times (-\frac{3}{4})^{n-1}.$$

2)

$$a = 1.5, \quad r = \frac{3}{1.5} = 2$$
$$u_n = 1.5 \times 2^{n-1}$$
$$u_{15} = 1.5 \times 2^{15-1} = 1.5 \times 2^{14} = 24576.$$

3)

$$a = 6, \quad r = \frac{8}{6} = \frac{4}{3}$$
$$u_n = 6 \times (\frac{4}{3})^{n-1}$$
$$u_6 = 6 \times (\frac{4}{3})^{6-1} = 6 \times (\frac{4}{3})^5 = \frac{2048}{81}.$$

4)

$$a = 0.004, \quad r = \frac{0.012}{0.004} = 3$$
$$u_n = 0.004 \times 3^{n-1}$$
$$57395.628 = 0.004 \times 3^{n-1}$$
$$\Rightarrow 3^{n-1} = 14348907$$
$$\Rightarrow \ln 3^{n-1} = \ln 14348907$$
$$\Rightarrow (n-1)\ln 3 = \ln 14348907$$
$$\Rightarrow n - 1 = \frac{\ln 14348907}{\ln 3}$$
$$\Rightarrow n = \frac{\ln 14348907}{\ln 3} + 1 = 16.$$

Alternative Method:

$$3^{n-1} = 14348907$$
$$\Rightarrow n - 1 = \log_3 14348907$$
$$\Rightarrow n = \log_3 14348907 + 1 = 16.$$

5)

$$\begin{aligned} a = 1.02, \quad r = \frac{4.08}{1.02} &= 4 \\ u_n &= 1.02 \times 4^{n-1} \\ 1069547.52 &= 1.02 \times 4^{n-1} \\ &\Rightarrow 4^{n-1} = 1048576 \\ &\Rightarrow \ln 4^{n-1} = \ln 1048576 \\ &\Rightarrow (n-1)\ln 4 = \ln 1048576 \\ &\Rightarrow n - 1 = \frac{\ln 1048576}{\ln 4} \\ &\Rightarrow n = \frac{\ln 1048576}{\ln 4} + 1 = 11. \end{aligned}$$

Alternative Method:

$$\begin{aligned} 4^{n-1} &= 1048576 \\ &\Rightarrow n - 1 = \log_4 1048576 \\ &\Rightarrow n = \log_4 1048576 + 1 = 11. \end{aligned}$$

6)

$$\begin{aligned} a = 10000, \quad r = \frac{11000}{10000} &= \frac{11}{10} \\ u_n &= 10000 \times (\frac{11}{10})^{n-1} \\ 21435.8881 &= 10000 \times (\frac{11}{10})^{n-1} \\ &\Rightarrow (\frac{11}{10})^{n-1} = 2.14358881 \\ &\Rightarrow \ln(\frac{11}{10})^{n-1} = \ln 2.14358881 \\ &\Rightarrow (n-1)\ln(\frac{11}{10}) = \ln 2.14358881 \\ &\Rightarrow n - 1 = \frac{\ln 2.14358881}{\ln(11/10)} \\ &\Rightarrow n = \frac{\ln 2.14358881}{\ln(11/10)} + 1 = 9. \end{aligned}$$

Alternative Method:

$$\begin{aligned} (\frac{11}{10})^{n-1} &= 2.14358881 \\ &\Rightarrow n - 1 = \log_{11/10} 2.14358881 \\ &\Rightarrow n = \log_{11/10} 2.14358881 + 1 = 9. \end{aligned}$$

7)

$$\begin{aligned}
a = 0.2, \quad r = 2 \\
u_n &= 0.2 \times 2^{n-1} \\
0.2 \times 2^{n-1} > 1000 &\Rightarrow 2^{n-1} > 5000 \\
&\Rightarrow \ln 2^{n-1} > \ln 5000 \Rightarrow (n-1)\ln 2 > \ln 5000 \\
&\Rightarrow n - 1 > \frac{\ln 5000}{\ln 2} \Rightarrow n > \frac{\ln 5000}{\ln 2} + 1 = 13.28\ldots \\
\therefore n &= 14 \\
u_{14} &= 1638.4.
\end{aligned}$$

8)

$$\begin{aligned}
u_4 = 28 \Rightarrow 28 = ar^{4-1} \Rightarrow ar^3 = 28 \quad (1) \\
u_{15} = 57344 \Rightarrow 57344 = ar^{15-1} \Rightarrow ar^{14} = 57344 \quad (2) \\
(2) \div (1): \quad \frac{ar^{14}}{ar^3} = \frac{57344}{28} \Rightarrow r^{11} = 2048 \\
\Rightarrow r = \sqrt[11]{2048} = 2 \\
\text{If } r = 2 \text{ then } a \times 2^3 = 28 \Rightarrow a = \frac{7}{2}.
\end{aligned}$$

9)

$$\begin{aligned}
u_3 = 2 \Rightarrow 2 = ar^{3-1} \Rightarrow ar^2 = 2 \quad (1) \\
u_7 = 13122 \Rightarrow 13122 = ar^{7-1} \Rightarrow ar^6 = 13122 \quad (2) \\
(2) \div (1): \quad \frac{ar^6}{ar^2} = \frac{13122}{2} \Rightarrow r^4 = 6561 \\
\Rightarrow r = \pm\sqrt[4]{6561} = \pm 9 \\
\text{If } r = 9 \text{ then } a \times 9^2 = 2 \Rightarrow a = \frac{2}{81} \\
\text{If } r = -9 \text{ then } a \times (-9)^2 = 2 \Rightarrow a = \frac{2}{81}.
\end{aligned}$$

10)

$$\begin{aligned}
u_2 = 15 \Rightarrow 15 = ar^{2-1} \Rightarrow ar = 15 \quad (1) \\
u_5 = -234375 \Rightarrow -234375 = ar^{5-1} \Rightarrow ar^4 = -234375 \quad (2) \\
(2) \div (1): \quad \frac{ar^4}{ar} = \frac{-234375}{15} \Rightarrow r^3 = -15625 \\
\Rightarrow r = \sqrt[3]{-15625} = -25 \\
\text{If } r = -25 \text{ then } a \times (-25) = 15 \Rightarrow a = -\frac{3}{5}.
\end{aligned}$$

11)

$$u_6 = 62.208 \Rightarrow 62.208 = ar^{6-1} \Rightarrow ar^5 = 62.208 \quad (1)$$
$$u_7 = 373.248 \Rightarrow 373.248 = ar^{7-1} \Rightarrow ar^6 = 373.248 \quad (2)$$
$$(2) \div (1): \quad \frac{ar^6}{ar^5} = \frac{373.248}{62.208} \Rightarrow r = 6$$
If $r = 6$ then $a \times 6^5 = 62.208 \Rightarrow a = 0.008$.

12)

$$u_3 = 19.8 \Rightarrow 19.8 = ar^{3-1} \Rightarrow ar^2 = 19.8 \quad (1)$$
$$u_6 = 534.6 \Rightarrow 534.6 = ar^{6-1} \Rightarrow ar^5 = 534.6 \quad (2)$$
$$(2) \div (1): \quad \frac{ar^5}{ar^2} = \frac{534.6}{19.8} \Rightarrow r^3 = 27 \Rightarrow r = 3$$
$$\text{When } r = 3, \quad a \times 3^2 = 19.8 \Rightarrow a = 2.2$$
$$\therefore u_n = 2.2 \times 3^{n-1}$$
$$2.2 \times 3^{n-1} > 10000000 \Rightarrow 3^{n-1} > \frac{50000000}{11}$$
$$\Rightarrow \ln 3^{n-1} > \ln \frac{50000000}{11}$$
$$\Rightarrow (n-1)\ln 3 > \ln \frac{50000000}{11}$$
$$\Rightarrow n - 1 > \frac{\ln(50000000/11)}{\ln 3}$$
$$\Rightarrow n > \frac{\ln(50000000/11)}{\ln 3} + 1 = 14.95\ldots$$
$$\text{So } n = 15$$
$$\therefore u_{15} = 10522531.8.$$

13 a)

$$u_2 = 1 + 2\sqrt{5} \Rightarrow 1 + 2\sqrt{5} = ar^{2-1}$$
$$\Rightarrow ar = 1 + 2\sqrt{5} \quad (1)$$
$$u_3 = 38 + 19\sqrt{5} \Rightarrow 38 + 19\sqrt{5} = ar^{3-1}$$
$$\Rightarrow ar^2 = 38 + 19\sqrt{5} \quad (2)$$
$$(2) \div (1): \quad \frac{ar^2}{ar} = \frac{38 + 19\sqrt{5}}{1 + 2\sqrt{5}} \Rightarrow r = \frac{38 + 19\sqrt{5}}{1 + 2\sqrt{5}}$$
$$\Rightarrow r = \frac{38 + 19\sqrt{5}}{1 + 2\sqrt{5}} \times \frac{1 - 2\sqrt{5}}{1 - 2\sqrt{5}}$$
$$= \frac{38 - 76\sqrt{5} + 19\sqrt{5} - 190}{1 - 2\sqrt{5} + 2\sqrt{5} - 20}$$
$$= \frac{-152 - 57\sqrt{5}}{-19} = \frac{-152}{-19} + \frac{-57\sqrt{5}}{-19} = 8 + 3\sqrt{5}.$$

13 b)

$$\begin{aligned} ar = 1+2\sqrt{5} &\Rightarrow a(8+3\sqrt{5}) = 1+2\sqrt{5} \\ &\Rightarrow a = \frac{1+2\sqrt{5}}{8+3\sqrt{5}} \Rightarrow a = \frac{1+2\sqrt{5}}{8+3\sqrt{5}} \times \frac{8-3\sqrt{5}}{8-3\sqrt{5}} \\ &= \frac{8-3\sqrt{5}+16\sqrt{5}-30}{64-24\sqrt{5}+24\sqrt{5}-45} \\ &= \frac{-22+13\sqrt{5}}{19} = -\frac{22}{19}+\frac{13}{19}\sqrt{5}. \end{aligned}$$

14)

$$\begin{aligned} u_1 = k+1, \quad u_2 = 5k, \quad u_3 = \frac{50k}{7} \\ r = \frac{u_2}{u_1} = \frac{u_3}{u_2} \Rightarrow \frac{5k}{k+1} = \frac{50k/7}{5k} \\ \Rightarrow \frac{5k}{k+1} = \frac{10}{7} \Rightarrow \frac{35k}{k+1} = 10 \\ \Rightarrow 35k = 10k+10 \Rightarrow 25k = 10 \Rightarrow k = \frac{10}{25} = \frac{2}{5}. \end{aligned}$$

15)

$$\begin{aligned} u_1 = \frac{49k}{36}, \quad u_2 = k+2, \quad u_3 = 3k-2 \\ r = \frac{u_2}{u_1} = \frac{u_3}{u_2} \Rightarrow \frac{k+2}{49k/36} = \frac{3k-2}{k+2} \Rightarrow \frac{36(k+2)}{49k} = \frac{3k-2}{k+2} \\ \Rightarrow 36(k+2)^2 = 49k(3k-2) \\ \Rightarrow 36(k^2+4k+4) = 147k^2 - 98k \\ \Rightarrow 36k^2+144k+144 = 147k^2 - 98k \\ \Rightarrow 0 = 111k^2 - 242k - 144 \\ \Rightarrow (37k+18)(3k-8) = 0 \Rightarrow k = -\frac{18}{37} \text{ or } k = \frac{8}{3}. \end{aligned}$$

16)

$$\begin{aligned} u_1 = 2^{3k-7}, \quad u_2 = 2^{2k+1}, \quad u_3 = 2^{4k-5} \\ r = \frac{u_2}{u_1} = \frac{u_3}{u_2} \Rightarrow \frac{2^{2k+1}}{2^{3k-7}} = \frac{2^{4k-5}}{2^{2k+1}} \\ \Rightarrow 2^{(2k+1)-(3k-7)} = 2^{(4k-5)-(2k+1)} \\ \Rightarrow 2^{-k+8} = 2^{2k-6} \Rightarrow -k+8 = 2k-6 \\ \Rightarrow 3k = 14 \Rightarrow k = \frac{14}{3}. \end{aligned}$$

17)

$$u_1 = 3e^{7/4+2k}, \quad u_2 = \frac{15}{2}e^{k+67/24}, \quad u_3 = \frac{75}{4}e^{101/24-k}$$

$$r = \frac{u_2}{u_1} = \frac{u_3}{u_2} \Rightarrow \frac{(15/2)e^{k+67/24}}{3e^{7/4+2k}} = \frac{(75/4)e^{101/24-k}}{(15/2)e^{k+67/24}}$$

$$\Rightarrow \frac{(5/2)e^{k+67/24}}{e^{7/4+2k}} = \frac{(5/2)e^{101/24-k}}{e^{k+67/24}}$$

$$\Rightarrow \frac{5}{2}e^{(k+67/24)-(7/4+2k)} = \frac{5}{2}e^{(101/24-k)-(k+67/24)}$$

$$\Rightarrow e^{-k+25/24} = e^{-2k+17/12} \Rightarrow -k + \frac{25}{24} = -2k + \frac{17}{12}$$

$$\Rightarrow k = \frac{3}{8}.$$

18)

$$a = 5, \quad r = \frac{10}{5} = 2$$

$$u_n = 5 \times 2^{n-1}$$

$$2560 = 5 \times 2^{n-1} \Rightarrow 2^{n-1} = 512$$

$$\Rightarrow \ln 2^{n-1} = \ln 512 \Rightarrow (n-1)\ln 2 = \ln 512$$

$$\Rightarrow n - 1 = \frac{\ln 512}{\ln 2} \Rightarrow n = \frac{\ln 512}{\ln 2} + 1 = 10$$

$$S_{10} = \frac{5(1-2^{10})}{1-2} = 5115.$$

Alternative Method:

$$2^{n-1} = 512 \Rightarrow n - 1 = \log_2 512$$

$$\Rightarrow n = \log_2 512 + 1 = 10$$

$$S_{10} = \frac{5(1-2^{10})}{1-2} = 5115.$$

19)

$$a = 0.13, \quad r = \frac{0.52}{0.13} = 4$$

$$u_n = 0.13 \times 4^{n-1}$$

$$2129.92 = 0.13 \times 4^{n-1} \Rightarrow 4^{n-1} = 16384$$

$$\Rightarrow \ln 4^{n-1} = \ln 16384 \Rightarrow (n-1)\ln 4 = \ln 16384$$

$$\Rightarrow n - 1 = \frac{\ln 16384}{\ln 4} \Rightarrow n = \frac{\ln 16384}{\ln 4} + 1 = 8$$

$$S_8 = \frac{0.13(1-4^8)}{1-4} = 2839.85.$$

Alternative Method:

$$4^{n-1} = 16384 \Rightarrow n - 1 = \log_4 16384$$

$$\Rightarrow n = \log_4 16384 + 1 = 8$$

$$S_8 = \frac{0.13(1-4^8)}{1-4} = 2839.85.$$

20)

$$a = \frac{1}{5000}, \quad r = \frac{1/1250}{1/5000} = 4$$
$$u_n = \frac{1}{5000} \times 4^{n-1}$$
$$\frac{8192}{625} = \frac{1}{5000} \times 4^{n-1} \Rightarrow 4^{n-1} = 65536$$
$$\Rightarrow \ln 4^{n-1} = \ln 65536 \Rightarrow (n-1)\ln 4 = \ln 65536$$
$$\Rightarrow n - 1 = \frac{\ln 65536}{\ln 4} \Rightarrow n = \frac{\ln 65536}{\ln 4} + 1 = 9$$
$$S_9 = \frac{\frac{1}{5000}(1 - 4^9)}{1 - 4} = 17.4762.$$

Alternative Method:

$$4^{n-1} = 65536 \Rightarrow n - 1 = \log_4 65536$$
$$\Rightarrow n = \log_4 65536 + 1 = 9$$
$$S_9 = \frac{\frac{1}{5000}(1 - 4^9)}{1 - 4} = 17.4762.$$

21)

$$a = 9000, \quad r = -\frac{9900}{9000} = -\frac{11}{10}$$
$$u_n = 9000 \times (-\frac{11}{10})^{n-1}$$
$$-17538.4539 = 9000 \times (-\frac{11}{10})^{n-1}$$
$$\Rightarrow (-\frac{11}{10})^{n-1} = -1.9487171$$
$$\Rightarrow (-1)^{n-1}(\frac{11}{10})^{n-1} = -1.9487171.$$

$(-1)^{n-1}$ must be equal to -1 (i.e. $n-1$ must be odd) as the right-hand side of the equation is negative:

$$-(\frac{11}{10})^{n-1} = -1.9487171 \Rightarrow (\frac{11}{10})^{n-1} = 1.9487171$$
$$\Rightarrow \ln(\frac{11}{10})^{n-1} = \ln 1.9487171$$
$$\Rightarrow (n-1)\ln\frac{11}{10} = \ln 1.9487171$$
$$\Rightarrow n - 1 = \frac{\ln 1.9487171}{\ln(11/10)}$$
$$\Rightarrow n = \frac{\ln 1.9487171}{\ln(11/10)} + 1 = 8$$
$$S_8 = \frac{9000(1 - (-11/10)^8)}{1 - (-11/10)} = -4901.0949.$$

Alternative Method:

$$(\frac{11}{10})^{n-1} = 1.9487171 \Rightarrow n - 1 = \log_{11/10} 1.9487171$$
$$\Rightarrow n = \log_{11/10} 1.9487171 + 1 = 8$$
$$S_8 = \frac{9000(1 - (-11/10)^8)}{1 - (-11/10)} = -4901.0949.$$

22)

$$\begin{aligned} a = \frac{2}{3}, \quad r = \frac{-1}{2/3} = -\frac{3}{2} \\ u_n = \frac{2}{3} \times (-\frac{3}{2})^{n-1} \\ \frac{19683}{512} = \frac{2}{3} \times (-\frac{3}{2})^{n-1} \\ \Rightarrow (-\frac{3}{2})^{n-1} = \frac{59049}{1024} \\ \Rightarrow (-1)^{n-1}(\frac{3}{2})^{n-1} = \frac{59049}{1024}. \end{aligned}$$

$(-1)^{n-1}$ must be equal to 1 (i.e. $n-1$ must be even) as the right-hand side of the equation is positive:

$$\begin{aligned} (\frac{3}{2})^{n-1} = \frac{59049}{1024} \Rightarrow \ln(\frac{3}{2})^{n-1} = \ln\frac{59049}{1024} \\ \Rightarrow (n-1)\ln\frac{3}{2} = \ln\frac{59049}{1024} \\ \Rightarrow n-1 = \frac{\ln(59049/1024)}{\ln(3/2)} \\ \Rightarrow n = \frac{\ln(59049/1024)}{\ln(3/2)} + 1 = 11 \\ S_{11} = \frac{\frac{2}{3}(1-(-3/2)^{11})}{1-(-3/2)} = \frac{35839}{1536}. \end{aligned}$$

Alternative Method:

$$\begin{aligned} (\frac{3}{2})^{n-1} = \frac{59049}{1024} \Rightarrow n-1 = \log_{3/2}\frac{59049}{1024} \\ \Rightarrow n = \log_{3/2}\frac{59049}{1024} + 1 = 11 \\ S_{11} = \frac{\frac{2}{3}(1-(-3/2)^{11})}{1-(-3/2)} = \frac{35839}{1536}. \end{aligned}$$

23 a)

$$\begin{aligned} a = 15, \quad r = 0.92 \\ u_n = 15 \times 0.92^{n-1} \\ u_7 = 15 \times 0.92^{7-1} = 15 \times 0.92^6 \\ = 9.10 \text{ miles to 2 d.p.} \end{aligned}$$

23 b)

$$S_7 = \frac{15(1-0.92^7)}{1-0.92} = 82.90 \text{ miles to 2 d.p.}$$

24 a)

$$a = \frac{4}{3} \times (\frac{2}{5})^{6-1} = \frac{4}{3} \times (\frac{2}{5})^5 = \frac{128}{9375}, \quad r = \frac{2}{5}.$$

24 b)

$$25 - 6 + 1 = 20 \text{ terms.}$$

25)

$$a = 66, \quad r = \frac{88}{66} = \frac{4}{3}$$
$$S_8 = \frac{66(1 - (4/3)^8)}{1 - 4/3} = 1779.77 \text{ to 2 d.p.}$$

26)

$$a = 5200, \quad r = \frac{4160}{5200} = \frac{4}{5}$$
$$S_9 = \frac{5200(1 - (4/5)^9)}{1 - 4/5} = 22510.34 \text{ to 2 d.p.}$$

27)

$$a = 128, \quad r = \frac{7}{2}$$
$$S_5 = \frac{128(1 - 16807/32)}{-5/2} = -\frac{256}{5} \times -\frac{16775}{32} = 26840.$$

28)

$$a = 19683, \quad r = \frac{1}{3}$$
$$S_{10} - S_3 = \frac{19683(1 - (1/3)^{10})}{1 - 1/3} - \frac{19683(1 - (1/3)^3)}{1 - 1/3}$$
$$= 29524 - 28431 = 1093.$$

29)

$$S_4 = 8S_2 \Rightarrow \frac{a(1 - r^4)}{1 - r} = \frac{8a(1 - r^2)}{1 - r}$$
$$\Rightarrow 1 - r^4 = 8(1 - r^2) \Rightarrow 1 - r^4 = 8 - 8r^2$$
$$\Rightarrow r^4 - 8r^2 + 7 = 0 \Rightarrow (r^2 - 7)(r^2 - 1) = 0$$
$$\Rightarrow r^2 = 7 \text{ or } r^2 = 1$$
$$\therefore r = \pm\sqrt{7} \text{ or } r = \pm 1.$$

As $r > 1$, we have that $r = \sqrt{7}$.

30 a)

$$a = 100, \quad r = \frac{9}{10}$$

$$\begin{aligned} S_k > 870 &\Rightarrow \frac{100(1-(9/10)^k)}{1-9/10} > 870 \\ &\Rightarrow 1-(\frac{9}{10})^k > \frac{87}{100} \\ &\Rightarrow (\frac{9}{10})^k < \frac{13}{100} \\ &\Rightarrow \ln(\frac{9}{10})^k < \ln\frac{13}{100} \\ &\Rightarrow k\ln\frac{9}{10} < \ln\frac{13}{100} \\ &\Rightarrow k > \frac{\ln(13/100)}{\ln(9/10)} = \frac{\ln 0.13}{\ln 0.9}. \end{aligned}$$

30 b)

$$\begin{aligned} k &> \frac{\ln 0.13}{\ln 0.9} = 19.36\ldots \\ \therefore k &= 20. \end{aligned}$$

31 a)

$$a = 2, \quad r = \frac{26}{25}$$

$$\begin{aligned} S_k > 1000 &\Rightarrow \frac{2(1-(26/25)^k)}{1-26/25} > 1000 \\ &\Rightarrow 1-(\frac{26}{25})^k < -20 \\ &\Rightarrow (\frac{26}{25})^k > 21 \\ &\Rightarrow \ln(\frac{26}{25})^k > \ln 21 \\ &\Rightarrow k\ln\frac{26}{25} > \ln 21 \\ &\Rightarrow k > \frac{\ln 21}{\ln(26/25)} = \frac{\ln 21}{\ln 1.04}. \end{aligned}$$

31 b)

$$\begin{aligned} k &> \frac{\ln 21}{\ln 1.04} = 77.62\ldots \\ \therefore k &= 78. \end{aligned}$$

32 a)

$$a = 20, \quad r = \frac{1}{8}$$

$$\begin{aligned} S_k < 22.85 &\Rightarrow \frac{20(1-(1/8)^k)}{1-1/8} < 22.85 \\ &\Rightarrow 1 - (\frac{1}{8})^k < \frac{3199}{3200} \\ &\Rightarrow (\frac{1}{8})^k > \frac{1}{3200} \\ &\Rightarrow \ln(\frac{1}{8})^k > \ln\frac{1}{3200} \\ &\Rightarrow k\ln\frac{1}{8} > \ln\frac{1}{3200} \\ &\Rightarrow k < \frac{\ln(1/3200)}{\ln(1/8)} = \frac{\ln 3200^{-1}}{\ln 8^{-1}} = \frac{-\ln 3200}{-\ln 8} = \frac{\ln 3200}{\ln 8}. \end{aligned}$$

32 b)

$$\begin{aligned} k &< \frac{\ln 3200}{\ln 8} = 3.88\ldots \\ \therefore k &= 3. \end{aligned}$$

33)

$$a = 5, \quad r = \frac{10}{5} = 2$$

$$\begin{aligned} S_k > 1500 &\Rightarrow \frac{5(1-2^k)}{1-2} > 1500 \\ &\Rightarrow 1 - 2^k < -300 \Rightarrow 2^k > 301 \\ &\Rightarrow \ln 2^k > \ln 301 \Rightarrow k\ln 2 > \ln 301 \\ &\Rightarrow k > \frac{\ln 301}{\ln 2} = 8.23\ldots \\ \therefore k &= 9. \end{aligned}$$

34)

$$a = 1.1, \quad r = \frac{3.3}{1.1} = 3$$

$$\begin{aligned} S_k > 30000 &\Rightarrow \frac{1.1(1-3^k)}{1-3} > 30000 \\ &\Rightarrow 1 - 3^k < -\frac{600000}{11} \\ &\Rightarrow 3^k > \frac{600011}{11} \\ &\Rightarrow \ln 3^k > \ln\frac{600011}{11} \\ &\Rightarrow k\ln 3 > \ln\frac{600011}{11} \\ &\Rightarrow k > \frac{\ln(600011/11)}{\ln 3} = 9.92\ldots \\ \therefore k &= 10. \end{aligned}$$

35 a)

$$\begin{aligned} a &= 250, \quad r = 1.03 \\ u_n &= 250 \times 1.03^{n-1} \\ u_5 &= 250 \times 1.03^{5-1} = 250 \times 1.03^4 = \text{£}281.38 \text{ to the nearest penny.} \end{aligned}$$

35 b)

$$\begin{aligned} 1000 = 250 \times 1.03^{n-1} &\Rightarrow 1.03^{n-1} = 4 \\ &\Rightarrow \ln 1.03^{n-1} = \ln 4 \Rightarrow (n-1)\ln 1.03 = \ln 4 \\ &\Rightarrow n - 1 = \frac{\ln 4}{\ln 1.03} \Rightarrow n = \frac{\ln 4}{\ln 1.03} + 1 = 47.8995\ldots \end{aligned}$$

So after 48 weeks.

35 c)

$$S_{52} = \frac{250(1 - 1.03^{52})}{1 - 1.03} = \text{£}30424.05 \text{ to the nearest penny.}$$

36)

$$r = \frac{14}{7} = 2.$$

As $r > 1$, the series is not convergent, and hence will have no sum to infinity.

37 a)

$$r = \frac{1.2}{0.4} = 3.$$

As $r > 1$, the series is not convergent.

37 b)

$$r = \frac{5}{10} = \frac{1}{2}.$$

As $-1 < r < 1$, the series is convergent.

37 c)

$$r = \frac{30}{20} = \frac{3}{2}.$$

As $r > 1$, the series is not convergent.

37 d)

$$r = \frac{95}{100} = \frac{19}{20}.$$

As $-1 < r < 1$, the series is convergent.

37 e)

$$r = -\frac{95}{100} = -\frac{19}{20}.$$

As $-1 < r < 1$, the series is convergent.

37 f)

$$r = -\frac{1.44}{1.6} = -0.9.$$

As $-1 < r < 1$, the series is convergent.

38 a)

$$a = 30, \quad r = 0.1$$
$$S_\infty = \frac{30}{1-(0.1)} = \frac{100}{3}.$$

38 b)

$$a = 800, \quad r = \frac{2}{5}$$
$$S_\infty = \frac{800}{1-(\frac{2}{5})} = \frac{4000}{3}.$$

38 c)

$$a = 99, \quad r = 0.99$$
$$S_\infty = \frac{99}{1-(0.99)} = 9900.$$

38 d)

$$a = -8000, \quad r = \frac{3}{8}$$
$$S_\infty = \frac{-8000}{1-(\frac{3}{8})} = -12800.$$

38 e)

$$a = 410, \quad r = -\frac{5}{6}$$
$$S_\infty = \frac{410}{1-(-\frac{5}{6})} = \frac{2460}{11}.$$

38 f)

$$a = -9, \quad r = -\frac{11}{12}$$
$$S_\infty = \frac{-9}{1-(-\frac{11}{12})} = -\frac{108}{23}.$$

39)

$$a = 80, \quad S_\infty = 90$$
$$\therefore 90 = \frac{80}{1-r} \Rightarrow 90(1-r) = 80 \Rightarrow 1 - r = \frac{8}{9}$$
$$\Rightarrow r = 1 - \frac{8}{9} = \frac{1}{9}.$$

40)

$$a = 1, \quad r = \frac{1}{3}$$
$$S_\infty = \frac{1}{1-1/3} = \frac{3}{2}.$$

41)

$$a = 100, \quad r = \frac{80}{100} = \frac{4}{5}$$
$$S_\infty = \frac{100}{1-4/5} = 500.$$

42)

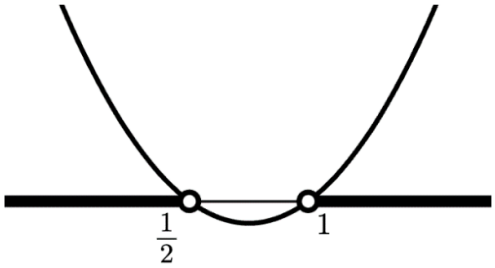
$$a = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}, \quad r = \frac{1/\sqrt{6}}{1/\sqrt{2}} = \frac{\sqrt{3}}{3}$$
$$S_\infty = \frac{\sqrt{2}/2}{1-\sqrt{3}/3} = \frac{3\sqrt{2}}{6-2\sqrt{3}}$$
$$= \frac{3\sqrt{2}}{6-2\sqrt{3}} \times \frac{6+2\sqrt{3}}{6+2\sqrt{3}} = \frac{18\sqrt{2}+6\sqrt{6}}{24}$$
$$= \frac{3\sqrt{2}+\sqrt{6}}{4}.$$

43)

$$a = 50, \quad S_\infty < 100$$
$$\frac{50}{1-r} < 100.$$

Multiply both sides by $(1-r)^2$ (to avoid multiplying by a negative):

$$\begin{aligned}
\frac{50(1-r)^2}{1-r} &< 100(1-r)^2 \\
&\Rightarrow 50(1-r) < 100(1-r)^2 \\
&\Rightarrow 1-r < 2(1-r)^2 \\
&\Rightarrow 1-r < 2(1-2r+r^2) \\
&\Rightarrow 1-r < 2-4r+2r^2 \\
&\Rightarrow 2r^2-3r+1 > 0 \\
&\Rightarrow (2r-1)(r-1) > 0.
\end{aligned}$$



$r < \frac{1}{2}$ or $r > 1$.

But as S_∞ exists, we must have $-1 < r < 1$.

$\therefore -1 < r < \frac{1}{2}$ is the required range of possible values of r.

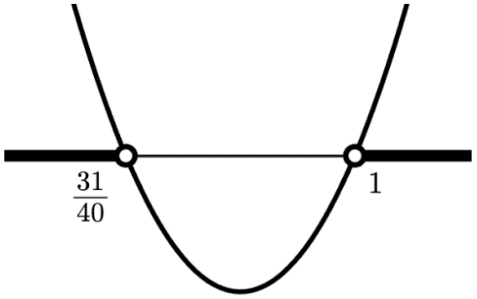
$$\{r : r \in \mathbb{R}, -1 < r < \tfrac{1}{2}\}.$$

44)

$$a = 9, \quad S_\infty < 40$$
$$\frac{9}{1-r} < 40.$$

Multiply both sides by $(1-r)^2$ (to avoid multiplying by a negative):

$$\begin{aligned}
\frac{9(1-r)^2}{1-r} &< 40(1-r)^2 \\
&\Rightarrow 9(1-r) < 40(1-r)^2 \\
&\Rightarrow 9 - 9r < 40(1 - 2r + r^2) \\
&\Rightarrow 9 - 9r < 40 - 80r + 40r^2 \\
&\Rightarrow 40r^2 - 71r + 31 > 0 \\
&\Rightarrow (40r - 31)(r - 1) > 0.
\end{aligned}$$



$r < \frac{31}{40}$ or $r > 1$.

But as S_∞ exists, we must have $-1 < r < 1$.

$\therefore -1 < r < \frac{31}{40}$ is the required range of possible values of r.

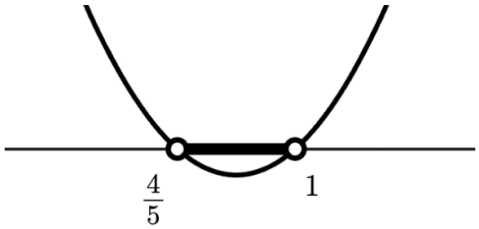
$$\{r : r \in \mathbb{R}, -1 < r < \tfrac{31}{40}\}.$$

45)

$$a = 100, \quad S_\infty > 500$$
$$\frac{100}{1-r} > 500.$$

Multiply both sides by $(1-r)^2$ (to avoid multiplying by a negative):

$$\begin{aligned}
\frac{100(1-r)^2}{1-r} &> 500(1-r)^2 \\
&\Rightarrow 100(1-r) > 500(1-r)^2 \\
&\Rightarrow 1-r > 5(1-r)^2 \\
&\Rightarrow 1-r > 5(1-2r+r^2) \\
&\Rightarrow 1-r > 5-10r+5r^2 \\
&\Rightarrow 5r^2-9r+4 < 0 \\
&\Rightarrow (5r-4)(r-1) < 0.
\end{aligned}$$



$\frac{4}{5} < r < 1$ is the required range of possible values of r.

$$\{r : r \in \mathbb{R}, \tfrac{4}{5} < r < 1\}.$$

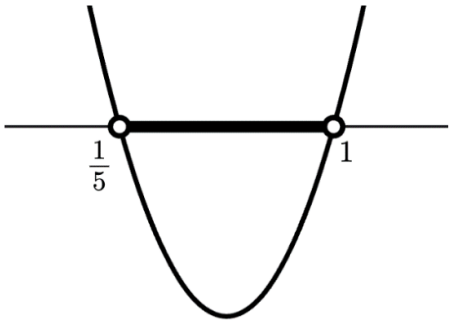
46)

$$a = 80, \quad 100 < S_\infty < 200$$

$$\frac{80}{1-r} > 100.$$

Multiply both sides by $(1-r)^2$ (to avoid multiplying by a negative):

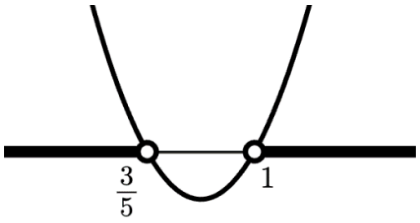
$$\begin{aligned}
\frac{80(1-r)^2}{1-r} &> 100(1-r)^2 \\
&\Rightarrow 80(1-r) > 100(1-r)^2 \\
&\Rightarrow 4(1-r) > 5(1-r)^2 \\
&\Rightarrow 4 - 4r > 5(1 - 2r + r^2) \\
&\Rightarrow 4 - 4r > 5 - 10r + 5r^2 \\
&\Rightarrow 5r^2 - 6r + 1 < 0 \\
&\Rightarrow (5r-1)(r-1) < 0.
\end{aligned}$$



$\frac{1}{5} < r < 1$.

Now solve $\frac{80}{1-r} < 200$:

$$\begin{aligned}
\frac{80(1-r)^2}{1-r} &< 200(1-r)^2 \\
&\Rightarrow 80(1-r) < 200(1-r)^2 \\
&\Rightarrow 2(1-r) < 5(1-r)^2 \\
&\Rightarrow 2 - 2r < 5(1 - 2r + r^2) \\
&\Rightarrow 2 - 2r < 5 - 10r + 5r^2 \\
&\Rightarrow 5r^2 - 8r + 3 > 0 \\
&\Rightarrow (5r-3)(r-1) > 0.
\end{aligned}$$



$r < \frac{3}{5}$ or $r > 1$.

$\therefore \frac{1}{5} < r < \frac{3}{5}$ is the required range of possible values of r.

$$\{r : r \in \mathbb{R}, \tfrac{1}{5} < r < \tfrac{3}{5}\}.$$

47)

$$a = 3000, \quad r = \frac{2}{3}$$
$$S_\infty = \frac{3000}{1 - 2/3} = 9000.$$

48)

$$a = 25350 \times (\frac{11}{13})^3 = \frac{199650}{13}, \quad r = \frac{11}{13}$$
$$S_\infty = \frac{199650/13}{1 - 11/13} = 99825.$$

49)

$$a = 1000000 \times (\frac{1}{5})^9 = \frac{64}{125}, \quad r = \frac{1}{5}$$
$$S_\infty = \frac{64/125}{1 - 1/5} = \frac{16}{25}.$$

50 a)

$$a = 18, \quad r = \frac{27/2}{18} = \frac{3}{4}$$
$$S_\infty = \frac{18}{1 - 3/4} = 72.$$

50 b)

$$v_1 = 18 \times 32 = 576$$
$$v_2 = \frac{27}{2} \times 32 = 432$$
$$v_3 = \frac{81}{8} \times 32 = 324$$
$$v_4 = \frac{243}{32} \times 32 = 243.$$

50 c)

$$\sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} 32u_n = 32\sum_{n=1}^{\infty} u_n = 32 \times 72 = 2304.$$

51)

$$\begin{aligned} u_2 = 10 \Rightarrow 10 = ar^{2-1} \Rightarrow ar = 10 \\ S_\infty = 40 \Rightarrow \frac{a}{1-r} = 40 \Rightarrow a = 40(1-r) \\ \therefore 40(1-r)r = 10 \Rightarrow (1-r)r = \frac{1}{4} \Rightarrow r - r^2 = \frac{1}{4} \\ \Rightarrow r^2 - r + \frac{1}{4} = 0 \Rightarrow (r - \frac{1}{2})^2 = 0 \Rightarrow r = \frac{1}{2} \\ a \times \frac{1}{2} = 10 \Rightarrow a = 20. \end{aligned}$$

52)

$$\begin{aligned} u_2 = 10800 \Rightarrow 10800 = ar^{2-1} \Rightarrow ar = 10800 \\ S_\infty = 45000 \Rightarrow \frac{a}{1-r} = 45000 \Rightarrow a = 45000(1-r) \\ \therefore 45000(1-r)r = 10800 \Rightarrow (1-r)r = \frac{6}{25} \Rightarrow r - r^2 = \frac{6}{25} \\ \Rightarrow r^2 - r + \frac{6}{25} = 0 \Rightarrow r = \frac{2}{5} \text{ or } r = \frac{3}{5} \\ \text{If } r = \frac{2}{5}, \text{ then } a = 45000(1 - \frac{2}{5}) = 27000 \\ \text{If } r = \frac{3}{5}, \text{ then } a = 45000(1 - \frac{3}{5}) = 18000. \end{aligned}$$

53)

$$\begin{aligned} u_2 = 95 \Rightarrow 95 = ar^{2-1} \Rightarrow ar = 95 \\ S_\infty = \frac{9500}{9} \Rightarrow \frac{a}{1-r} = \frac{9500}{9} \Rightarrow a = \frac{9500}{9}(1-r) \\ \therefore \frac{9500}{9}(1-r)r = 95 \Rightarrow (1-r)r = \frac{9}{100} \Rightarrow r - r^2 = \frac{9}{100} \\ \Rightarrow r^2 - r + \frac{9}{100} = 0 \Rightarrow r = \frac{1}{10} \text{ or } r = \frac{9}{10}. \end{aligned}$$

It is given that $r > \frac{1}{2}$ so $r = \frac{9}{10}$.

$$\begin{aligned} \text{If } r = \frac{9}{10} \text{ then } a \times \frac{9}{10} = 95 \Rightarrow a = \frac{950}{9} \\ S_6 = \frac{(950/9)(1-(9/10)^6)}{1-9/10} = \frac{8902621}{18000} = 494.59 \text{ to 2 d.p.} \end{aligned}$$

54)

$$\frac{a}{1-r} = 10 \times \frac{a(1-r^2)}{1-r} \Rightarrow \frac{1}{1-r} = \frac{10(1-r^2)}{1-r}$$
$$\Rightarrow 1 = 10(1-r^2) \Rightarrow \frac{1}{10} = 1-r^2 \Rightarrow r^2 = \frac{9}{10}$$
$$\Rightarrow r = \pm\frac{3\sqrt{10}}{10}.$$

55)

$$u_1 = \frac{5}{k}, \quad u_2 = \frac{1}{k^4}, \quad u_3 = \frac{k}{80}$$
$$r = \frac{u_2}{u_1} = \frac{u_3}{u_2} \Rightarrow \frac{1/k^4}{5/k} = \frac{k/80}{1/k^4} \Rightarrow \frac{1}{k^8} = \frac{1}{16}$$
$$\Rightarrow k^8 = 16 \Rightarrow k = \pm\sqrt{2}.$$

k is a positive constant so $k = \sqrt{2}$.

$$a = \frac{5}{\sqrt{2}}, \quad r = \frac{1/(\sqrt{2})^4}{5/\sqrt{2}} = \frac{1}{4} \times \frac{\sqrt{2}}{5} = \frac{\sqrt{2}}{20}$$
$$S_\infty = \frac{5/\sqrt{2}}{1-\sqrt{2}/20} = \frac{5}{\sqrt{2}-1/10} = \frac{50}{10\sqrt{2}-1}$$
$$= \frac{50}{10\sqrt{2}-1} \times \frac{10\sqrt{2}+1}{10\sqrt{2}+1}$$
$$= \frac{500\sqrt{2}+50}{199} = \frac{500}{199}\sqrt{2} + \frac{50}{199}.$$

56)

$$a = 10, \quad r = \frac{10/(2k)}{10} = \frac{1}{2k}.$$

In order for a sum to infinity to exist, $-1 < r < 1$, so $-1 < \frac{1}{2k} < 1 \Rightarrow -2 < \frac{1}{k} < 2$.

Consider $-2 < \frac{1}{k}$, this is true for all positive constants k, so does not carry any restrictions.

Now consider $\frac{1}{k} < 2 \Rightarrow 1 < 2k \Rightarrow k > \frac{1}{2}$.

$\therefore k > \frac{1}{2}$.

57)

$$a = (\frac{1}{2}\ln e^2 - 2\ln e^{1/4})^1 = \frac{1}{2} \times 2 - 2 \times \frac{1}{4} = 1 - \frac{1}{2} = \frac{1}{2}$$
$$r = \frac{1}{2}\ln e^2 - 2\ln e^{1/4} = \frac{1}{2}$$
$$S_\infty = \frac{1/2}{1-1/2} = 1.$$

58)

$$\begin{aligned}
a &= (\sin 45^\circ \cos 30^\circ)^1 = \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2\sqrt{2}} \\
&= \frac{\sqrt{3}}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{6}}{4} \\
r &= \sin 45^\circ \cos 30^\circ = \frac{\sqrt{6}}{4} \\
S_\infty &= \frac{\sqrt{6}/4}{1-\sqrt{6}/4} = \frac{\sqrt{6}}{4-\sqrt{6}} \\
&= \frac{\sqrt{6}}{4-\sqrt{6}} \times \frac{4+\sqrt{6}}{4+\sqrt{6}} = \frac{4\sqrt{6}+6}{10} = \frac{2\sqrt{6}+3}{5}.
\end{aligned}$$

59)

$$\begin{aligned}
a = u_1 &= (\frac{\sin(90+180\times 1)^\circ}{5})^1 = (-\frac{1}{5})^1 = -\frac{1}{5} \\
u_2 &= (\frac{\sin(90+180\times 2)^\circ}{5})^2 = (\frac{1}{5})^2 = \frac{1}{25} \\
r &= \frac{1/25}{-1/5} = -\frac{1}{5} \\
S_\infty &= \frac{-1/5}{1-(-1/5)} = -\frac{1/5}{6/5} = -\frac{1}{6}.
\end{aligned}$$

60)

$$\begin{aligned}
a &= (\frac{\sqrt{5}}{2} - \frac{\sqrt{3}}{3})^1 = \frac{3\sqrt{5}-2\sqrt{3}}{6} \\
r &= \frac{\sqrt{5}}{2} - \frac{\sqrt{3}}{3} = \frac{3\sqrt{5}-2\sqrt{3}}{6} \\
S_\infty &= \frac{(3\sqrt{5}-2\sqrt{3})/6}{1-(3\sqrt{5}-2\sqrt{3})/6} \\
&= \frac{3\sqrt{5}-2\sqrt{3}}{6-(3\sqrt{5}-2\sqrt{3})} = \frac{3\sqrt{5}-2\sqrt{3}}{6-3\sqrt{5}+2\sqrt{3}} \\
&= \frac{3\sqrt{5}-2\sqrt{3}}{6-3\sqrt{5}+2\sqrt{3}} \times \frac{6-3\sqrt{5}-2\sqrt{3}}{6-3\sqrt{5}-2\sqrt{3}} \\
&= \frac{-33+18\sqrt{5}-12\sqrt{3}}{69-36\sqrt{5}} \\
&= \frac{-33+18\sqrt{5}-12\sqrt{3}}{69-36\sqrt{5}} \times \frac{69+36\sqrt{5}}{69+36\sqrt{5}} \\
&= \frac{963-432\sqrt{15}+54\sqrt{5}-828\sqrt{3}}{-1719} \\
&= -\frac{107}{191} + \frac{48}{191}\sqrt{15} - \frac{6}{191}\sqrt{5} + \frac{92}{191}\sqrt{3}.
\end{aligned}$$

61)

$$a = \frac{100}{9}, \quad r = \frac{3}{5}$$

$$\frac{100/9}{1-3/5} - \frac{(100/9)(1-(3/5)^k)}{1-3/5} > 0.09$$

$$\Rightarrow \frac{250}{9} - \frac{250}{9}(1-(\frac{3}{5})^k) > 0.09$$
$$\Rightarrow \frac{250}{9}(1-(\frac{3}{5})^k) < \frac{24919}{900}$$
$$\Rightarrow 1-(\frac{3}{5})^k < \frac{24919}{25000}$$
$$\Rightarrow (\frac{3}{5})^k > \frac{81}{25000}$$
$$\Rightarrow \ln(\frac{3}{5})^k > \ln(\frac{81}{25000})$$
$$\Rightarrow k\ln\frac{3}{5} > \ln(\frac{81}{25000})$$
$$\Rightarrow k < \frac{\ln(81/25000)}{\ln(3/5)} = 11.22 \text{ to 2 d.p.}$$

62)

$$S = a + ar + ar^2 + \cdots + ar^{n-1} \quad (1)$$
$$rS = ar + ar^2 + ar^3 + \cdots + ar^n \quad (2)$$
$$(2)-(1): \quad rS - S = ar^n - a$$
$$\Rightarrow S(r-1) = a(r^n-1) \Rightarrow S = \frac{a(r^n-1)}{r-1}.$$

63)

$$a = 8000, \quad r = \frac{1}{2}$$

$$\frac{8000}{1-1/2} - \frac{8000(1-(1/2)^k)}{1-1/2} < 5000$$

$$\Rightarrow 16000 - 16000(1-(\frac{1}{2})^k) < 5000$$
$$\Rightarrow 16000(1-(\frac{1}{2})^k) > 11000$$
$$\Rightarrow 1-(\frac{1}{2})^k > \frac{11}{16}$$
$$\Rightarrow (\frac{1}{2})^k < \frac{5}{16}$$
$$\Rightarrow \ln(\frac{1}{2})^k < \ln\frac{5}{16}$$
$$\Rightarrow k\ln\frac{1}{2} < \ln\frac{5}{16}$$
$$\Rightarrow k > \frac{\ln(5/16)}{\ln(1/2)} = 1.68 \text{ to 2 d.p.}$$

64 a)

$$u_5 = k \times 0.5^{5-1} = k \times 0.5^4 = \frac{1}{16}k.$$

64 b)

$$a = k, \quad r = \frac{1}{2}$$
$$S_\infty = \frac{k}{1 - 1/2} = \frac{k}{1/2} = 2k.$$

64 c)

$$\begin{aligned}\sum_{n=1}^{\infty} v_n &= \sum_{n=1}^{\infty}(0.5 + k \times 0.5^{n-1}) \\ &= \sum_{n=1}^{\infty} 0.5 + \sum_{n=1}^{\infty} k \times 0.5^{n-1} \\ &= (\sum_{n=1}^{\infty} 0.5) + 2k.\end{aligned}$$

$\sum_{n=1}^{\infty} 0.5$ diverges as it is adding together an infinite number of 0.5s. Therefore, the sum to infinity of v_n does not converge.

65)

$$u_1 = 2\pi \times 40 = 80\pi$$
$$a = 80\pi, \quad r = 0.98$$
$$S_\infty = \frac{80\pi}{1 - 0.98} = 4000\pi.$$

So $k = 4000\pi$.

66)

$$a = 108, \quad r = \frac{3}{8}$$
$$\begin{aligned}u_n &= 108 \times (\frac{3}{8})^{n-1} = (2^2 \times 3^3) \times (\frac{3}{2^3})^{n-1} \\ &= (2^2 \times 3^3) \times \frac{3^{n-1}}{2^{3(n-1)}} = 2^{2-3(n-1)} \times 3^{3+(n-1)} \\ &= 2^{5-3n} \times 3^{2+n} = \frac{3^{2+n}}{2^{3n-5}}.\end{aligned}$$

67)

$$\begin{aligned} a &= 400, \quad r = \frac{5}{4} \\ u_n &= 400 \times (\frac{5}{4})^{n-1} = (2^4 \times 5^2) \times (\frac{5}{2^2})^{n-1} \\ &= (2^4 \times 5^2) \times \frac{5^{n-1}}{2^{2(n-1)}} = 2^{4-2(n-1)} \times 5^{2+(n-1)} \\ &= 2^{6-2n} \times 5^{1+n} = \frac{5^{n+1}}{2^{2n-6}}. \end{aligned}$$

68)

$$\begin{aligned} a = 2k^2 + 5k - 6, \quad r &= \frac{2k + 5 - 6/k}{2k^2 + 5k - 6} \\ &= \frac{(1/k)(2k^2 + 5k - 6)}{2k^2 + 5k - 6} = \frac{1}{k} \\ S_\infty = 3 \Rightarrow \frac{2k^2 + 5k - 6}{1 - 1/k} &= 3 \Rightarrow \frac{2k^3 + 5k^2 - 6k}{k - 1} = 3 \\ \Rightarrow 2k^3 + 5k^2 - 6k &= 3k - 3 \\ \Rightarrow 2k^3 + 5k^2 - 9k + 3 &= 0 \\ \Rightarrow (2k - 1)(k^2 + 3k - 3) = 0 \Rightarrow k &= \frac{1}{2} \text{ or } k = \frac{-3 \pm \sqrt{21}}{2}. \end{aligned}$$

Note that $-1 < \frac{1}{k} < 1$ so $k = \frac{1}{2}$ does not converge.

$k = \frac{-3+\sqrt{21}}{2}$ does not converge.

$k = \frac{-3-\sqrt{21}}{2}$ does converge.

So $k = \frac{-3-\sqrt{21}}{2}$.

69)

$$\begin{aligned} a = \frac{k+1}{2k+3}, \quad r &= \frac{k+1}{2k+3} \\ S_\infty = 2 \Rightarrow \frac{(k+1)/(2k+3)}{1 - (k+1)/(2k+3)} &= 2 \\ \Rightarrow \frac{k+1}{2k+3-(k+1)} = 2 \Rightarrow \frac{k+1}{k+2} &= 2 \\ \Rightarrow k + 1 = 2k + 4 \Rightarrow k &= -3 \\ \therefore a = \frac{-3+1}{2(-3)+3} = \frac{2}{3}, \quad r &= \frac{2}{3} \\ S_6 = \frac{(2/3)(1 - (2/3)^6)}{1 - 2/3} &= \frac{1330}{729}. \end{aligned}$$

70 a)

$$u_1 = 5, \quad u_2 = 8, \quad u_3 = 11, \quad u_4 = 14, \ldots$$

This is an arithmetic sequence, with first term 5 and common difference 3.

70 b)

$$u_1 = 4, \quad u_2 = 8, \quad u_3 = 16, \quad u_4 = 32, \ldots$$

This is a geometric sequence, with first term 4 and common ratio 2.

70 c)

$$u_1 = 6, \quad u_2 = 15, \quad u_3 = 33, \quad u_4 = 69, \ldots$$

This is not an arithmetic sequence as the difference between the terms is not constant: $15 - 6 = 9$, $33 - 15 = 18$, $69 - 33 = 36$, ...

This is not a geometric sequence as the ratio between each successive pair of terms is not constant: $\frac{15}{6} = \frac{5}{2}$, $\frac{33}{15} = \frac{11}{5}$, $\frac{69}{33} = \frac{23}{11}$, ...

The nth term for this sequence (beyond the specification) is $u_n = \frac{3}{2}(3 \times 2^n - 2)$.

71 a)

$$\begin{aligned}
u_1 = 3, \quad u_2 = 7, \quad u_3 = 11, \quad u_4 = 15, \ldots \\
u_n = 3 + (n-1) \times 4 \\
v_1 = 2, \quad v_2 = 6, \quad v_3 = 18, \quad v_4 = 54, \ldots \\
v_n = 2 \times 3^{n-1} \\
u_1 + v_1 = 5, \quad u_2 + v_2 = 13, \quad u_3 + v_3 = 29, \\
u_4 + v_4 = 69, \ldots \\
u_n + v_n = 3 + (n-1) \times 4 + 2 \times 3^{n-1}.
\end{aligned}$$

This is not an arithmetic sequence as the difference between the terms is not constant: $13 - 5 = 8$, $29 - 13 = 16$, $69 - 29 = 40$, ...

This is not a geometric sequence as the ratio between each successive pair of terms is not constant: $\frac{13}{5}$, $\frac{29}{13}$, $\frac{69}{29}$, ...

71 b)

$$\begin{aligned}
\sum_{n=1}^{8}(u_n + v_n) &= \sum_{n=1}^{8} u_n + \sum_{n=1}^{8} v_n \\
&= \sum_{n=1}^{8}(3 + (n-1) \times 4) + \sum_{n=1}^{8} 2 \times 3^{n-1} \\
&= \frac{1}{2} \times 8 \times (2 \times 3 + (8-1) \times 4) + \frac{2(1 - 3^8)}{1 - 3} \\
&= 136 + 6560 = 6696.
\end{aligned}$$

72 a)

Let the arithmetic sequence be u_n and the geometric sequence be v_n.

$$\begin{aligned}
\therefore u_n &= 10 + (n-1) \times 700 \\
\therefore v_n &= 5 \times 3^{n-1} \\
v_k > u_k &\Rightarrow 5 \times 3^{k-1} > 10 + (k-1) \times 700 \\
&\Rightarrow 5 \times 3^k \times 3^{-1} > 10 + 700k - 700 \\
&\Rightarrow \frac{5}{3} \times 3^k > 700k - 690 \\
&\Rightarrow 3^k > 420k - 414 \\
&\Rightarrow 3^k + 414 > 420k.
\end{aligned}$$

72 b)

When $k = 7$:

LHS $= 2601$, RHS $= 2940$.

When $k = 8$:

LHS $= 6975$, RHS $= 3360$.

So $k = 8$.